



C23-EC-ECII-402

23431

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – FOURTH SEMESTER EXAMINATION

MICROCONTROLLERS AND APPLICATIONS

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. List the interrupts of 8051 microcontroller and give their priority. 3
- 2. State the function of PSW register of 8051 microcontroller. 3
- 3. Distinguish between machine cycle and T-state. 3
- 4. Classify the instructions of 8051 based on their operation. 3
- 5. Define a subroutine and state its use. 2+1
- 6. Write PUSH and POP instructions of 8051 with an example for each. 3
- 7. Draw the circuit for interfacing of 8051 with relay to drive a lamp. 3
- 8. Write any six command codes of 16x2 LCD module. 3
- 9. List any three reasons for writing programs in embedded C. 3
- 10. Write an 8051 C program to toggle all the bits of P2 continuously. 3

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[Contd...

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** Draw the functional block diagram of 8051 microcontrollers and explain each block. 10
- 12.** Explain the modes of operations of serial communication with 8051. 10
- 13.** Classify and explain instructions of 8051 based on their length with examples. 10
- 14.** Explain the operation carried out during execution of the following instructions by assuming suitable data :
(a) MOVX A, @A+DPTR $2\frac{1}{2}+2\frac{1}{2}+2\frac{1}{2}+2\frac{1}{2}$
(b) RRCA
(c) DAA
(d) NOP
- 15.** Define debugging and explain about single step and break point debugging techniques. 10
- 16.** Write an Assembly Language program to add two 8-bit numbers. Assume that the numbers are stored in 4500H and 4501H locations. Store the result in memory location 4502H and store if any carry generated in 4503H. 10
- 17.** Interface Common Anode Seven Segment Display with 8051 and write a program to display decimal Numbers from 0 to 9. 10
- 18.** Write a C program for 8051 to transfer the letter "A" SERIALY AT 9600 baud continuously. Use 8-bit data and one stop bit. 10

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